

CTQ - Clinical Trial Qualifier

AI-Powered Clinical Trial Matching and Eligibility Analysis for Indian Patients

This report explains the complete, final system: what it does, the workflow pipeline, how every technology in the stack works (each with a worked example), a guided tour of the website with live screenshots of the current build, and the measured evaluation results of live runs from 29 Sep 2026.

One-line summary: you enter a patient (structured form / CSV upload / cohort mode, or unstructured clinical notes as PDF, photo, CSV, Excel, TXT or JSON), a DistilBERT ML NER turns free text into structured facts, CTQ screens that patient against every trial in its 84-trial CTRI/CTG database with semantic search + a deterministic rule engine + a capped Groq LLM second opinion, and returns one of three explained verdicts per trial: Potentially Eligible, Not Eligible, or Insufficient Information. A Model Evaluation page scores the whole pipeline against ground-truth labels, and a Trial Database page browses and live-imports trials from ClinicalTrials.gov.

Live measured results quoted throughout (29 Sep 2026): a structured sweep of 80 synthetic patients × 84 trials (1,600 labelled pairs) scored Accuracy 93.9% on the 1,454 decidable pairs (specificity 99.2%); the unstructured end-to-end run (10 notes → ML NER → matching → 17 labelled pairs) scored Accuracy 88.2% (Precision 80%, Recall 80%, F1 0.80). A 10-patient × 84-trial cohort completes in about 100 seconds after the parallel-extraction and LLM-cap performance work.

1. What the project does

Clinical trials publish eligibility criteria as long free-text lists: age windows, lab cut-offs ("HbA1c between 7.5 and 10.5 percent"), medication requirements ("on stable metformin for at least 3 months"), exclusion clauses. Patients and clinicians rarely have the time to read every trial's criteria. CTQ automates exactly that reading.

The system performs five jobs:

1. Understand the patient. Two input modes on one page: the **Structured Data** tab (60+ guided form fields, or a structured CSV upload, with a cohort mode for many patients) and the **Unstructured Data** tab (clinical notes as PDF, photo, CSV, Excel, TXT or JSON). Non-English notes (Hindi and 12 other scripts) are auto-translated to English before extraction. Free text becomes structured facts via the DistilBERT ML NER - no hand-written regex discovery.

2. Find candidate trials. The database holds 84 trials in CTRI format (32 synthetic CTRI + 52 live-imported from ClinicalTrials.gov, recruiting in India). A structured pre-filter annotates trials; the 'Trials to check' picker lets the user scope the run to All trials or the top-5/10/15 best matches.

3. Rank by meaning. The patient profile and each trial's criteria are embedded into vectors (all-MiniLM-L6-v2, local) and compared with cosine similarity. This is the Match Score % - 'how similar is this patient to this trial'.

4. Verify the details. A deterministic rule engine checks everything objectively checkable on EVERY trial: numeric lab cut-offs parsed from the criterion text, age and gender windows, medication requirements and durations, comorbidity exclusions, pregnancy and lifestyle criteria. Missing inclusion-side facts produce 'Insufficient Information' rather than a guess; missing exclusion-side facts keep 'Potentially Eligible' with the gap listed. On the top-15 most similar trials a Groq LLM adds an adversarial second opinion (never overriding rule vetoes).

5. Explain, import, export, and evaluate. Every verdict ships with itemised reasons quoting the trial's own criteria; single results download as CSV, the cohort exports as Excel; the Trial Database page imports

fresh trials from ClinicalTrials.gov in the UI; the Model Evaluation page scores the pipeline against ground-truth labels.

2. Workflow pipeline - one patient, end to end

The flowchart below is the methodology of the whole system; the numbered steps after it walk one real patient (P001, the diabetes sample note) through every stage with the actual data at each step.

CTQ Methodology — end-to-end matching pipeline

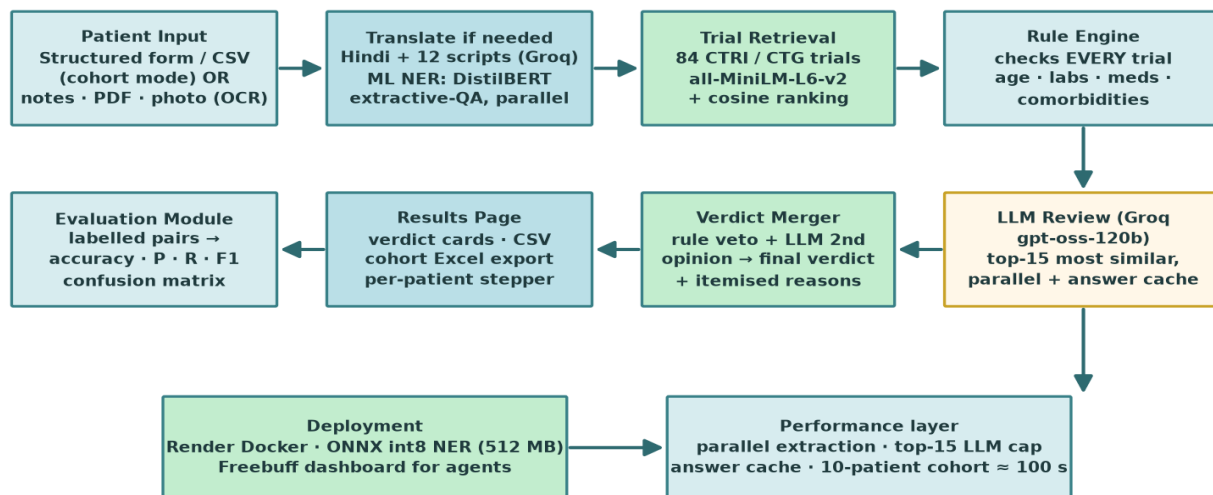


Figure 1 - CTQ end-to-end methodology flowchart (regenerated from the live build).

Step 1 - Input. P001 arrives either as form fields or as the note: “48-year-old male with Type 2 Diabetes Mellitus diagnosed 4 years ago. Height 170 cm, weight 82 kg. Increased thirst and fatigue. On stable Metformin 500 mg twice daily for the last 12 months. HbA1c 8.5 percent, eGFR 80, ALT 25 U/L. No history of liver disease. Does not smoke. Never drinks alcohol.”

Step 2 - Translation (conditional). services/translate.py scans the alphabetic characters against Unicode script ranges; $\geq 5\%$ non-Latin triggers a Groq medical translation (numbers, units and lab names preserved verbatim). P001 is pure English, so this stage is a pass-through; a Hindi note (■■■■■ 48 ■■■■) would be translated here and the UI would show a ■■ badge.

Step 3 - ML extraction. The DistilBERT extractive-QA NER asks targeted questions (“What is the patient’s HbA1c level?”) and normalises validated spans: age 48, male, Type 2 Diabetes Mellitus, duration 48 months, Metformin 500 mg twice daily (12 mo), HbA1c 8.5, eGFR 80, ALT 25, height 170 / weight 82 → BMI 28.4, Liver disease: No, smoking Never, alcohol Never. Multi-note files are extracted in parallel (6 workers) and the model load is lock-guarded.

Step 4 - Retrieval + ranking. build_profile() normalises the facts into a PatientProfile; all 84 trials’ criteria are embedded (cached) and the profile text is embedded once; cosine similarity ranks every trial and computes the Match Score %. A top-K scope may limit downstream checking to the K best matches.

Step 5 - Rule engine (every trial). For trial CTRI/2024/01/062001 (T2DM add-on to metformin): “Patient age 48 is within the required range 30-65 years”, “HbA1c 8.5 satisfies inclusion criterion ‘HbA1c between 7.5 and 10.5 percent’”, “Current medication Metformin matches trial requirement of stable metformin ≥ 3 months (12 months recorded)” → **Potentially Eligible**. Against the obesity trial CTRI/2025/08/064022: “BMI 28.4 does not satisfy inclusion criterion ‘Body mass index between 30 and 45’” → **Not Eligible**.

Step 6 - LLM review (top-15 only). The 15 most similar trials additionally go to Groq gpt-oss-120b as an adversarial second opinion (a parallel worker pool; successful answers are cached per patient+trial). It can flag a violation the parser missed or add narrative, but can never overturn a rule-engine hard failure. Trials outside the top-15 keep their rule-engine verdict unchanged.

Step 7 - Verdict merge + explanation. merge_rule_and_llm applies the deterministic policy: rule veto wins → Not Eligible; rule-detected missing inclusion facts → Insufficient Information; LLM-only concrete violation → Not Eligible; otherwise Potentially Eligible. Every verdict carries itemised reasons quoting the trial's criteria, the Match Score %, phase, sponsor and India locations.

Step 8 - Output. The Results page renders verdict cards (green/amber/red badges), supports per-patient stepping for cohorts, downloads a results CSV; the cohort exports one Excel file; the labelled evaluation path produces accuracy/precision/recall/F1 and the confusion matrix.

3. Why this design is good

Honest about uncertainty. Most systems force a yes/no answer. CTQ has a third verdict, 'Insufficient Information', and names exactly which fields are missing. In screening, a wrong confident answer is far more harmful than an honest 'need more data'. In the live structured sweep, 104 of 1,600 pairs returned Insufficient Information instead of a guess - and that honesty is tracked as its own metric (insufficient-detection rate 50% on labelled-insufficient pairs).

Auditable, not a black box. Every verdict lists the exact criteria checked (e.g. 'HbA1c 8.5 satisfies inclusion criterion "HbA1c between 7.5 and 10.5"'). Because structured facts are checked with code, the reasons are reproducible - the ML only extracts, it never invents facts and can never overturn a rule-engine hard failure.

ML where ML wins, rules where rules win. Language understanding (reading a note, coping with phrasing variety) is ML's strength; safety-critical verification (a numeric cut-off, a hard exclusion) is code's strength. CTQ assigns each to the right component, and the LLM is only a third, capped opinion.

Built for the Indian context. Trials are modelled on the Clinical Trials Registry - India (CTRI) plus live ClinicalTrials.gov recruiting-in-India imports, with city/state locations shown on every card, and Hindi/Indic notes supported end to end.

Fast enough for real cohorts. Note extraction runs in parallel threads with a lock-guarded model load; per-trial work runs in a thread pool; the LLM review is capped (CTQ_LLM_REVIEW_TOP_N, default 15) and answer-cached. A 10-patient × 84-trial cohort completes in ≈100 s on a laptop CPU; the rule engine keeps verdicts flowing even when the Groq quota is exhausted.

Measurable. The Model Evaluation page has a one-click bundled sample (10 notes + 17 labels) and accepts full uploads (80 patients × 84 trials shipped as backend/data/sample_*.csv): anyone can reproduce the metrics in this report in minutes.

4. How accurate is it - live measured results

4.1 Run A - structured sweep (80 patients × 84 trials, run 3, 29 Sep 2026)

80 fully structured synthetic patients (backend/data/sample_patients.csv) were labelled against all 84 trials (1,600 ground-truth pairs: 157 eligible, 1,401 not eligible, 42 insufficient). Predictions of 'Insufficient Information' (104 pairs) are excluded from the binary matrix and reported separately - the system refused to guess where data was missing:

Metric	Value
Labelled pairs	1,600 (1,454 scored binary; 104 predicted-Insufficient excluded; 42 labelled-Insufficient)

Accuracy	93.9%
Specificity	99.2% (only 10 false alarms across 1,313 not-eligible pairs)
Precision / Recall / F1 (eligible class)	86.1% / 44.0% / 0.58
Confusion matrix	TP 62 · FP 10 · FN 79 · TN 1,303

Reading it honestly: the system almost never wrongly rejects (specificity 99.2%) and rarely cries wolf (precision 86.1%), but its recall on the eligible class is conservative - where a profile is missing a fact a criterion needs, it returns Insufficient Information instead of forcing an 'Eligible'. Several trials score 100% individually (per-trial table on the Evaluation page); the misses concentrate in criteria that need facts the form never captured. This is the designed trade-off: a screening tool must prefer 'need more data' over a false 'yes'.

4.2 Run B - unstructured end-to-end (10 notes → ML NER → 17 labelled pairs)

The bundled sample ships as two CSVs (sample_data/evaluation_patients_unstructured.csv + evaluation_labels_ground_truth.csv) and also via the one-click 'Load bundled sample data' button. The full pipeline - including the ML extraction step - ran live:

Metric	Value
Labelled pairs	17 (0 excluded, 0 skipped)
Accuracy	88.2% (15 of 17 correct)
Precision / Recall / F1	80% / 80% / 0.80
Confusion matrix	TP 4 · FP 1 · FN 1 · TN 11

CTQ Evaluation Results

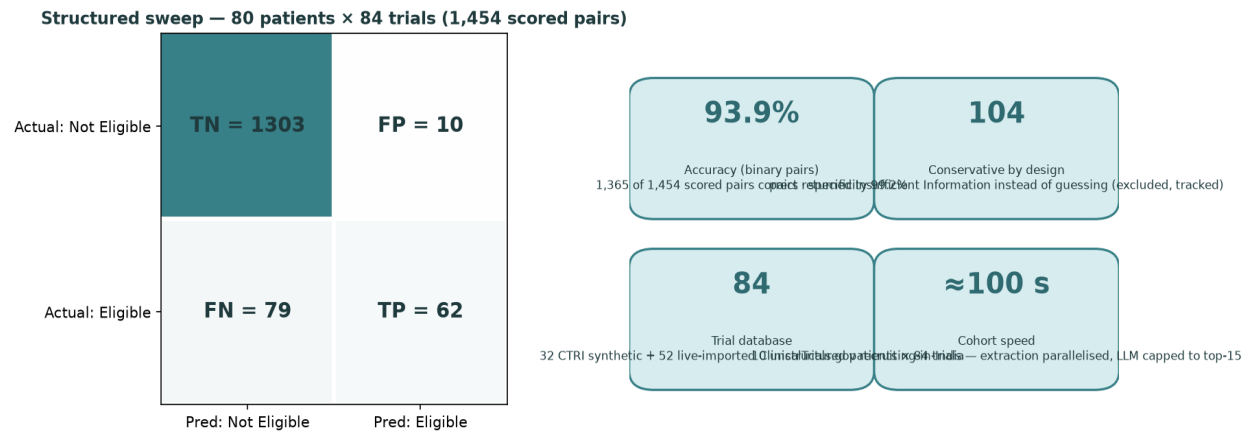


Figure 2 - Confusion matrix of Run A and headline tiles. Run B's smaller matrix (TP 4, FP 1, FN 1, TN 11) exercises the extra extraction step end to end: two pairs diverge from ground truth because the note text genuinely lacked (or phrased) a criterion-critical fact - exactly the honest-failure mode the design targets.

4.3 Ranking scope - accuracy against the top-K matches

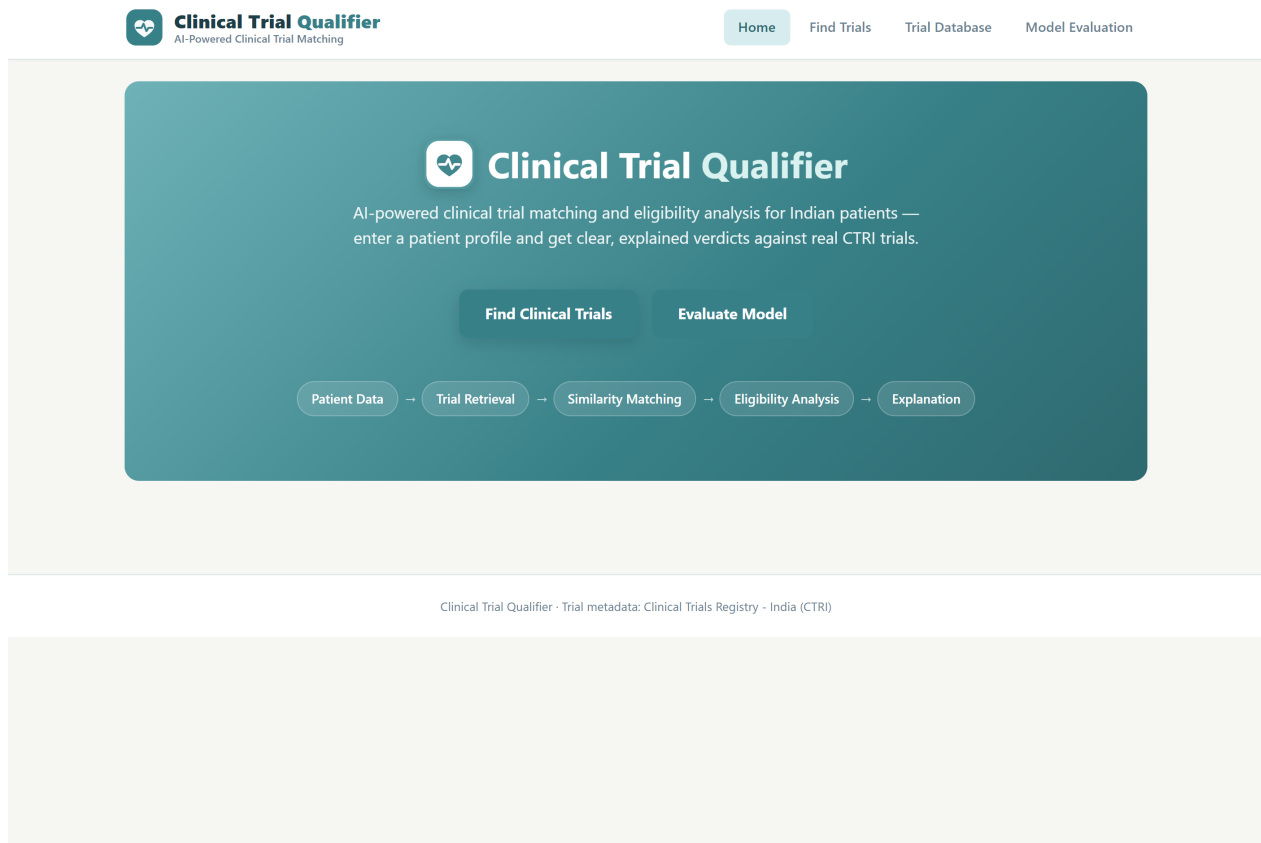
In real use, a coordinator reads the ranked list from the top - not all 84 trials. The Model Evaluation page therefore offers a ranking scope: All trials, Top 3, Top 5 or Top 10. Under a top-K scope, a labelled pair is scored only when that trial is among the patient's K best-matching trials (by embedding similarity); pairs outside the scope are skipped and reported next to the confusion matrix, so the denominator is always visible.

4.4 Honesty mechanism

Rows whose prediction is 'Insufficient Information' are excluded from the binary matrix and reported separately. If a note genuinely lacks a fact - e.g. an asthma note without FEV1 - the matching pair degrades to 'Insufficient Information' instead of a lucky guess, which is the behaviour a screening tool should have. Both evaluation datasets exist as standalone CSVs in `sample_data/` for manual upload.

5. Guided tour - how each part works (live screenshots)

5.1 Home - the front door



The hero states what the product does with two calls to action: Find Clinical Trials (the matcher) and Evaluate Model (accuracy measurement). The five pills show the pipeline a search will follow: Patient Data, Trial Retrieval, Similarity Matching, Eligibility Analysis, Explanation.

5.2 Find Trials - two input modes, one page

Patient → Trial Matcher

Enter patient information to find potentially relevant Indian clinical trials. All fields except age are optional - missing information leads to honest "Insufficient Information" verdicts rather than guesses.

Try it instantly with sample data

[Sample: Diabetes \(47F\)](#)
[Sample: Hypertension \(58M\)](#)
[Sample: Asthma \(24F\)](#)
[Sample: Healthy volunteer \(28M\)](#)

Single patient mode

Filling several patients? Add them below - each one gets its own trial check.

[+ Add another patient](#)
[↓ Download all patients \(Excel\)](#)
[↑ Upload structured CSV](#)
[Structured Data \(form\)](#)
[Unstructured Data \(photos, PDF, CSV\)](#)

Basic Information

Patient ID

Age *

Gender

State

Any code, e.g. P001 - no personal names needed

City

Height (cm)

Weight (kg)

Disease Information

Primary disease / condition

Disease duration (months)

Disease severity

Previous diagnosis

e.g. Type 2 Diabetes

Symptoms

☐ Increased thirst

☐ Frequent urination

☐ Fatigue

☐ Blurred vision

☐ Numbness in feet

The Structured Data tab is open by default: one-click sample profiles (Diabetes 47F, Hypertension 58M, Asthma 24F, Healthy volunteer 28M), the cohort bar ('Single patient mode' / '+ Add another patient' / 'Download all patients (Excel)' / 'Upload structured CSV'), and the Unstructured Data tab switch at the top. The 'Trials to check' picker (All / Top 5 / Top 10 / Top 15) sits with the submit button.

5.3 Structured form - medications and labs

Medications

[Remove](#)

[Remove](#)
[+ Add medication](#)

Previous treatment failure?

☐ Yes

☐ No

☒ Unknown

Known drug allergies

Did a prior therapy fail?

Laboratory Values (all optional)

HbA1c (%)

e.g. 8.2

Fasting glucose (mg/dL)

Systolic BP (mmHg)

Diastolic BP (mmHg)

Creatinine (mg/dL)

eGFR

Hemoglobin (g/dL)

WBC (cells/uL)

Platelets (/uL)

ALT (U/L)

AST (U/L)

Bilirubin (mg/dL)

Cholesterol (mg/dL)

FEV1 (% predicted)

respiratory trials, e.g. 78

Trials needing other measurements are handled via the notes fields below.

Other Eligibility Information

Pregnancy status

Smoking status

Alcohol use

Previous clinical-trial participation

The diabetes sample is loaded. Each medication row captures name, dose ('500 mg'), frequency ('twice daily') and duration in months - the details criteria like 'stable metformin for at least 3 months' need. The Laboratory Values grid holds HbA1c, glucose, BP, creatinine, eGFR, haemoglobin, WBC, platelets, ALT, AST, bilirubin, cholesterol and FEV1 % predicted. Left-blank fields become Unknown - which is exactly what produces honest 'Insufficient Information' verdicts instead of guesses.

5.4 Cohort mode - many patients, one run

**Clinical Trial Qualifier**
AI-Powered Clinical Trial Matching

HomeFind TrialsTrial DatabaseModel Evaluation

Patient → Trial Matcher

Enter patient information to find potentially relevant Indian clinical trials. All fields except age are optional - missing information leads to honest "Insufficient Information" verdicts rather than guesses.

Try it instantly with sample data

Sample: Diabetes (47F)Sample: Hypertension (58M)Sample: Asthma (24F)Sample: Healthy volunteer (28M)

2 patients in the cohort

Patient 1 is the form above; the others are listed below it.

+ Add another patient↓ Download all patients (Excel)↑ Upload structured CSV

Clear cohort (2 patients)

Structured Data (form)Unstructured Data (photos, PDF, CSV)

Patient 2 of 2

× Remove

Patient ID	Age *	Gender	Primary condition
P002		Select...	e.g. Type 2 Diabetes
Disease duration (months)	Height (cm)	Weight (kg)	HbA1c (%)
eGFR	FEV1 (% predicted)	Pregnancy status	Smoking status
		Select...	Select...
Alcohol use			
Select...			

After '+ Add another patient', each added patient gets its own full form card ('Patient 2 of 2'), the counter reads '2 patients in the cohort', and the submit button becomes 'Find Trials for all 2 patients'. Results come back per patient with a ← Previous / Next patient → stepper, and the whole cohort's entered details download as one Excel file. Structured CSV upload fills the same form + cohort states (patient 1's meds/symptoms land in the dedicated form states).

5.5 Results - 'Your Trial Matches'

Your Trial Matches

New search

84 trials · 30 potentially eligible · 49 not eligible · 5 need more info

Eligibility

Match score

All

Highest match score first

Disease

All

State

All

Recruitment status

All

Phase

All

CTRI/2024/01/062001

A Randomised Open Label Trial of Glimepiride Add-on to Metformin in Type 2 Diabetes Mellitus

✓ **Potentially Eligible** **Match Score: 78%** Type 2 Diabetes Mellitus Recruiting Phase III Siliguri, West Bengal

Why?

- ✓ Patient age 47 is within the required range 30-65 years
- ✓ Patient condition "Type 2 Diabetes" matches trial condition "Type 2 Diabetes Mellitus"
- ✓ HbA1c 8.2 satisfies inclusion criterion "HbA1c between 7.5 and 10.5 percent at screening"
- ✓ BMI 29.7 satisfies inclusion criterion "Body mass index between 23 and 35"
- ✓ Patient is on metformin, satisfying "On stable metformin therapy for"
- ✓ No conflict with the trial's pregnancy/breastfeeding exclusion
- ✓ Medical history shows no liver disease, matching exclusion about liver disease
- ✓ Age 47 within required 30-65 years
- ✓ HbA1c 8.2% within required 7.5-10.5% range

Missing information

- ✗ eGFR value to assess renal function
- ✗ ALT level to assess hepatic impairment

After pressing Find Matching Trials, the backend runs the pipeline and the Results page lists every trial with its verdict badge (green Potentially Eligible, red Not Eligible, amber Insufficient Information), Match Score %, registry ID, locations and phase, plus summary counts. A 'Download results (CSV)' button sits below the list.

5.6 Every verdict is explained

- ✓ Patient does not take insulin; no conflict with exclusion "Current use of insulin"
- ✓ No conflict with the trial's pregnancy/breastfeeding exclusion
- ✓ Medical history shows no kidney disease, matching exclusion about kidney disease / renal impairment
- ✗ Disease duration 24 months exceeds "newly diagnosed within 6 month(s)" (trial seeks newly diagnosed patients)

View Full Trial

NCT07165028

A Master Protocol of Multiple Agents in Adults With Metabolic Dysfunction-Associated Steatotic Liver Disease (SYNERGY-Outcomes)

✓ **Potentially Eligible** **Match Score: 58%** Metabolic Dysfunction-Associated Steatotic Liver Disease Recruiting Phase3 Ahmedabad · Bhubaneswar ...

Why?

- ✓ Patient age 47 is within the required range >= 18 years
 - ✓ HbA1c 8.2 does not trigger exclusion "" Have a hemoglobin A1c (HbA1c) greater than 10%"
- No known exclusion criterion detected.

View Full Trial

NCT07613307

A Study of Orforglipron (LY3502970) in Participants With Type 2 Diabetes Who Observe Ramadan Fasting

✓ **Potentially Eligible** **Match Score: 57%** Diabetes Mellitus, Type 2 Recruiting Phase3 Agra · Hyderabad ...

Why?

- ✓ Patient age 47 is within the required range 18-65 years
- ✓ Patient condition "Type 2 Diabetes" matches trial condition "Diabetes Mellitus, Type 2"
- ✓ HbA1c 8.2 satisfies inclusion criterion "" Have an HbA1c value of at least 7.0% (53 millimoles per mole (mmol/mol)) to less than 9.5% (91 mmol/mol) at screening."
- ✓ Medical history shows no cancer, matching exclusion about active cancer
- ✓ Medical history shows no autoimmune disease, matching exclusion about autoimmune disease

Each card carries green reasons-for (e.g. 'HbA1c 8.5 satisfies inclusion criterion "HbA1c between 7.5 and 10.5 percent"'), red reasons-against (e.g. 'BMI 28.4 does not satisfy "Body mass index between 30 and 45"'), and amber missing-information naming the exact fields never provided. This is the anti-black-box guarantee: the same list a clinician would produce by hand, generated in seconds.

5.7 Unstructured Data tab - drop any document

**Clinical Trial Qualifier**
AI-Powered Clinical Trial Matching

HomeFind TrialsTrial DatabaseModel Evaluation

Patient → Trial Matcher

Enter patient information to find potentially relevant Indian clinical trials. All fields except age are optional - missing information leads to honest "Insufficient Information" verdicts rather than guesses.

Try it instantly with sample data

Sample: Diabetes (47F)

Sample: Hypertension (58M)

Sample: Asthma (24F)

Sample: Healthy volunteer (28M)

Structured Data (form)

Unstructured Data (photos, PDF, CSV)

Unstructured patient data

Upload a photo of a prescription or discharge summary (JPG/PNG), a PDF, or a CSV/Excel/TXT file of medical history. The DistilBERT ML NER reads the text, converts it into structured profile fields, and every trial in the database is checked.

Choose file

No file chosen

Clinical Trial Qualifier · Trial metadata: Clinical Trials Registry - India (CTRI)

Accepts PDF, photos (OCR via RapidOCR), CSV, Excel, TSV, TXT and JSON. A text column is found by alias (notes / clinical_notes / summary / medical_history...) and a patient id column by alias (patient_id / note_id / id). Multi-note CSVs show a per-patient stepper after extraction.

5.8 Ten notes extracted - live table

Patient → Trial Matcher

Enter patient information to find potentially relevant Indian clinical trials. All fields except age are optional - missing information leads to honest "Insufficient Information" verdicts rather than guesses.

Try it instantly with sample data

Sample: Diabetes (47F)

Sample: Hypertension (58M)

Sample: Asthma (24F)

Sample: Healthy volunteer (28M)

Structured Data (form)

Unstructured Data (photos, PDF, CSV)

Unstructured patient data

Upload a photo of a prescription or discharge summary (JPG/PNG), a PDF, or a CSV/Excel/TXT file of medical history. The DistilBERT ML NER reads the text, converts it into structured profile fields, and every trial in the database is checked.

Choose file unstructured_...patients.csv

Extracted structured data from 5 notes in "unstructured_5_patients.csv" (table).

D001

Patient 1 of 5

Age 48

Male

Type 2 Diabetes Mellitus

48 mo

Smoke: Never

Alcohol: Never

N-A

Metformin

Increased thirst

Fatigue

hba1c 8.5

egfr 80

alt 25

Trials to check:

All trials

Top 5

Top 10

Top 15

Find Matching Trials for this profile

Find Trials for all 5 patients

← Previous patient

Patient 1 / 5

Next patient →

Download extracted results (CSV)

The uploaded 10-note CSV ('U-patients_find_trials.csv' in this capture) is converted by the DistilBERT NER into structured facts per patient - here U001: age 48, male, Type 2 Diabetes Mellitus, 48 months, Metformin, increased thirst, fatigue, HbA1c 8.5, eGFR 80, ALT 25. Multi-note extraction runs six notes in parallel (≈ 270 s sequential $\rightarrow \approx 72$ s), then 'Find Trials for all 10 patients' screens all 84 trials per patient in ≈ 100 s.

5.9 Trial Database - browse and live-import

Trial Database

Every trial CTQ checks patients against — the bundled CTRI-format demo set plus real, currently-recruiting studies ingested live from ClinicalTrials.gov (India sites). Click a row to see its full criteria.

Live ingestion — import recruiting trials from ClinicalTrials.gov

Fetches real studies from the ClinicalTrials.gov API v2, maps them onto CTQ's trial schema (criteria, phase, age windows, India sites) and adds them to the database — they immediately flow through the same matching pipeline. Imports are kept across restarts.

Anaemia ☒ India sites only ☒ Recruiting only

Search Condition

Status Phase Type

84 shown · 84 in database

TRIAL ID	TITLE	STATUS	PHASE	GENDER	AGE	SITES	SOURCE
CTRI/2024/01/062001	A Randomised Open Label Trial of Glimepiride Add-on to Metformin in Type 2 Diabetes Mellitus	Recruiting	Phase III	Both	30–65	Siliguri, West Bengal	CTRI demo
CTRI/2024/02/062115	Observational Study of Glycaemic Variability in Newly Diagnosed Type 2 Diabetes Patients in Urban India	Recruiting	N/A	Both	25–60	Bengaluru, Karnataka · Madurai, Tamil Nadu ...	CTRI demo
CTRI/2024/03/062318	Evaluation of a Fixed Dose Combination of Perindopril and Indapamide in Uncontrolled Hypertension	Recruiting	Phase III	Both	40–75	Amritsar, Punjab	CTRI demo
CTRI/2024/04/062402	Phase II Study of a Novel HER2 Targeted Therapy in Metastatic Breast Cancer	Recruiting	Phase II	Female	18–70	Kolkata, West Bengal	CTRI demo

The dedicated page lists all 84 trials with search and filters (condition, status, phase, study type). The import panel pulls fresh studies from ClinicalTrials.gov's API v2 by condition preset (Type 2 Diabetes, Hypertension, Asthma, COPD, ...), optionally India-only and Recruiting-only, maps them into the CTRI schema, and merges them into the running database (persisted in a sidecar JSON and replayed on restart).

5.10 Model Evaluation - zero-preparation demo + ranking scope

Model Evaluation

Give **unstructured patient data** (notes / summaries / CSV of medical history) plus **labelled structured data** (ground truth). The ML NER converts each unstructured patient into a structured profile, every trial in the database is checked, and the predictions are scored against your labels: accuracy, precision, recall, F1 and the confusion matrix.

Sample data

No files handy? Load the synthetic sample set shipped with the app: 10 patient notes plus 17 labelled patient/trial pairs. Then just press **Run Evaluation**.

[Load bundled sample data \(10 patients, 17 labels\)](#)

1 · Unstructured patient data * (PDF / photo / CSV / Excel / TXT / JSON)

A file with clinical notes per patient — e.g. columns like `patient_id`, `notes` (CSV), one note per line (TXT), or a document of medical history. Accepted formats:

`.pdf .png .jpg .jpeg .csv .xlsx .tsv .txt .json`

[Choose file](#) No file chosen

2 · Labelled structured data * (ground truth)

CSV with columns `patient_id`, `trial_id`, `actual_label` — actual_label: 1 = Potentially Eligible, 0 = Not Eligible, 2 = Insufficient Information.

[Choose file](#) No file chosen

3 · Ranking scope

Which trials should accuracy be scored against? **All trials** checks every trial in the database. **Top-K** scores only each patient's K best-matching trials (by match score) — the way a coordinator actually reads the ranked list. Labelled pairs that fall outside a patient's top-K are skipped and reported.

[All trials](#)

[Top 3 trials](#)

[Top 5 trials](#)

[Top 10 trials](#)

4 · Run evaluation

[Run Evaluation](#)

Upload both files to enable evaluation.

Four numbered steps: unstructured patient file, labelled ground-truth file, ranking scope, run. The 'Load bundled sample data (10 patients, 21 labels)' button fills both inputs from the app itself - no file handling needed for a demo. The ranking-scope buttons choose between scoring all trials and scoring only each patient's top-3 / top-5 / top-10 best matches.

5.11 The live metrics dashboard

Model Evaluation

Give **unstructured patient data** (notes / summaries / CSV of medical history) plus **labelled structured data** (ground truth). The ML NER converts each unstructured patient into a structured profile, every trial in the database is checked, and the predictions are scored against your labels: accuracy, precision, recall, F1 and the confusion matrix.

Sample data

No files handy? Load the synthetic sample set shipped with the app: 10 patient notes plus 21 labelled patient/trial pairs. Then just press **Run Evaluation**.

[Load bundled sample data \(10 patients, 21 labels\)](#)

1 · Unstructured patient data * (PDF / photo / CSV / Excel / TXT / JSON)

A file with clinical notes per patient — e.g. columns like `patient_id`, `notes` (CSV), one note per line (TXT), or a document of medical history. Accepted formats:

`.pdf .png .jpg .jpeg .csv .xlsx .tsv .txt .json`

[Choose file](#) No file chosen

Loaded **10** patient notes from `evaluation_patients_unstructured.csv` (sample).

2 · Labelled structured data * (ground truth)

CSV with columns `patient_id`, `trial_id`, `actual_label` — actual_label: 1 = Potentially Eligible, 0 = Not Eligible, 2 = Insufficient Information.

[Choose file](#) No file chosen

Loaded **17** labels from `evaluation_labels_ground_truth.csv`.

3 · Ranking scope

Which trials should accuracy be scored against? **All trials** checks every trial in the database. **Top-K** scores only each patient's K best-matching trials (by match score) — the way a coordinator actually reads the ranked list. Labelled pairs that fall outside a patient's top-K are skipped and reported.

[All trials](#)

[Top 3 trials](#)

[Top 5 trials](#)

[Top 10 trials](#)

A real run: metric tiles for Accuracy / Precision / Recall / F1, the colour-coded confusion matrix, the skipped/excluded pair counts printed beside it, and an expandable 'ML extraction details' panel listing the facts the NER found for each patient. Evaluation runs are stored in SQLite and browsable under previous runs.

5.12 Per-pair predictions vs labels



Clinical Trial Qualifier
AI-Powered Clinical Trial Matching

[Home](#)
[Find Trials](#)
[Trial Database](#)
[Model Evaluation](#)

Predicted: Eligible

4

TP

1

FN

Predicted: Not Eligible

1

FP

11

TN

TP = true positive, FP = false positive, FN = false negative, TN = true negative. Rows where the model answered "Insufficient Information" are excluded from the binary matrix and reported separately.

▶ ML extraction details (10 patients)

Predictions vs labels

PATIENT	TRIAL	PREDICTED	ACTUAL	MATCH
P008	CTRI/2024/06/062633	Potentially Eligible	Eligible	✓
P008	CTRI/2024/01/062001	Potentially Eligible	Eligible	✓
P009	CTRI/2025/04/063673	Potentially Eligible	Eligible	✓
P004	CTRI/2025/04/063673	Not Eligible	Not Eligible	✓
P010	CTRI/2024/11/063128	Potentially Eligible	Eligible	✓
P001	CTRI/2024/11/063128	Not Eligible	Not Eligible	✓
P006	CTRI/2024/09/062966	Not Eligible	Not Eligible	✓
P004	CTRI/2024/05/062517	Not Eligible	Not Eligible	✓
P003	CTRI/2024/06/062633	Not Eligible	Not Eligible	✓
P004	CTRI/2024/03/062318	Not Eligible	Not Eligible	✓
P009	CTRI/2024/08/062851	Not Eligible	Not Eligible	✓

Each scored row with a ✓: prediction equals the ground-truth label. The table downloads as CSV ('evaluation_predictions.csv') for offline inspection.

6. The tech stack - what each technology does and how

The app is a classic two-part web application: a React single-page app in the browser talking to a Python FastAPI server. The table summarises every piece; the paragraphs after explain the interesting ones with a worked example each.

Technology	Role in CTQ	How it is used here
React 18 + Vite + react-router	Frontend UI	Five routes: Home, Find Trials (structured/unstructured tabs, cohort, top-K picker), Results, Trial Data
FastAPI + Uvicorn	Backend REST API	/api endpoints: patient/analyze (top_k), cohort/analyze + cohort/export (Excel), extract/document, pati
Pydantic	Data models + validation	PatientProfile, Medication, Trial and result schemas; coerces form/CSV values and guarantees every
DistilBERT extractive-QA (distilbert-base-uncased-distilled-squad)	ML NER inside distilbert	Asks one targeted question per clinical field per note ('What is the patient's HbA1c level?'); picks the b
Groq API (openai/gpt-oss-120b)	Translation + LLM review	Two jobs: (1) medical translation of non-Latin notes (Hindi + 12 scripts) to English before extraction; (2)
sentence-transformers (all-MiniLM-L6-v2)	384-d embeddings	Encodes the patient profile text and each trial's criteria into 384-d vectors (exact-text cache, batched);
Custom rule engine (services/eligibility.py)	Domain-specific checking	Parses numeric cut-offs from criterion text ('HbA1c between 7.5 and 10.5', 'eGFR below 45', '3 × ULN
RapidOCR (ONNX) + PyMuPDF	OCR of PDF reading	Photos/scans of prescriptions → RapidOCR text; scanned PDF pages are rasterised with PyMuPDF t
pandas + openpyxl	Tabular parsing / Excel	Reads uploaded CSV/TSV/Excel with column aliasing; openpyxl writes the cohort Excel export.
SQLite	Local storage	Stores evaluation runs; trials.json (84 trials, 200+ criteria) plus the live-import sidecar JSON loads at s
ReportLab + matplotlib	This report & figures	The PDF you are reading is generated from live numbers; matplotlib renders the confusion matrix and
Playwright	Testing & screenshots	Automated end-to-end browser runs produced every screenshot in section 5 and regression-check the
Render (Docker)	Hosting	Auto-deploys on every push to master; ONNX int8 NER artifact fits the 512 MB free container; free-tie

6.1 The ML NER step in detail - with the worked example

Input note (P001): “48-year-old male with Type 2 Diabetes Mellitus diagnosed 4 years ago. Height 170 cm, weight 82 kg. Increased thirst and fatigue. On stable Metformin 500 mg twice daily for the last 12 months. HbA1c 8.5 percent, eGFR 80, ALT 25 U/L. No history of liver disease. Does not smoke.” The engine asks the model a list of targeted questions and keeps a span only when it passes validation:

“What is the patient's HbA1c level?” → span “8.5” - accepted because the number sits next to the words 'HbA1c' (anchor check, word-boundary matched so 'alt' never matches inside 'heALThy').

“Does the patient have any history of liver disease?” → “No history of liver disease” → normalised to comorbidity 'Liver disease: No' - which the rule engine later uses to satisfy that trial's exclusion clause.

The diagnosis question initially returned “liver disease” from the negated sentence; a negation guard rejects spans found inside “No history of...” contexts, and a sentence-level fallback then lifts the real diagnosis 'Type 2 Diabetes Mellitus'.

Numbers already claimed by one field (height 170) cannot be stolen by another (BP), preventing cross-contaminated extractions; blood pressure must match a real '150/95' digit-slash-digit pair so 'ALT 26 U/L' cannot be read as a BP of 26.

Output facts: age 48, gender Male, condition Type 2 Diabetes Mellitus, duration 48 months, Metformin 500 mg twice daily (12 mo), HbA1c 8.5, eGFR 80, ALT 25, height/weight → BMI 28.4, Liver disease: No, smoking Never.

Parallelism. A 10-note upload is extracted by 6 threads (≈270 s → ≈72 s; results are byte-identical to sequential extraction - verified). A module lock guarantees exactly one model copy is loaded even under concurrency.

6.2 The rule engine in detail - where safety lives

6.3 The multilingual translation layer (Hindi today, 13 scripts)

Detect. `services/translate.py` scans the note's alphabetic characters against Unicode script ranges (Devanagari, Bengali, Gurmukhi, Gujarati, Odia, Tamil, Telugu, Kannada, Malayalam, CJK, Cyrillic, Arabic, Hebrew, Greek). If $\geq 5\%$ belong to a non-Latin script, the note is flagged; detection is instant and offline.

Extract. The English text flows through the untouched, validated NER pipeline - no change to extraction behaviour.

Hindi-verified result: age 48, male, diabetes, Metformin 500 mg twice daily, HbA1c 8.5, creatinine 1.1, eGFR 80, BP 140/85, Never/Never lifestyle - the complete profile, from pure Hindi input.

6.4 The performance architecture (new)

Per-trial parallelism. Rule-engine checks and LLM calls for the ranked trials run in a `ThreadPoolExecutor` (the regex parsing and the Groq HTTP call release the GIL). One patient $\times$ 84 trials no longer walks a for-loop.

Answer cache + breaker. Successful LLM answers are cached per (patient text, trial id), so re-running a cohort or evaluation is instant; the rate-limit breaker trips on Groq 429s and falls back to rule-engine-only within seconds instead of stalling in retry backoffs.

Measured effect. 10 notes: ≈ 270 s $\rightarrow \approx 72$ s. 10-patient $\times$ 84-trial cohort with matching: ≈ 100 s end to end, all 10 patients, complete verdicts.

6.5 The evaluation pipeline in detail

POST /api/patient/analyze-batch receives the unstructured patients and the labels; each note is re-extracted by the ML NER (explicit fields and pre-extracted facts win), trial criteria vectors are batch-embedded once, patient vectors in one batch, and the rule-engine checks run in a thread pool. Labelled pairs are scored: 1 = eligible, 0 = not eligible, 2 = insufficient (excluded from the binary matrix, tracked separately). Per-trial and per-patient accuracy breakdowns, the extraction log and a downloadable predictions CSV come back with the metrics. The frontend's 'Load bundled sample data' button simply calls /api/evaluation/sample-data, which serves exactly these CSVs.

6.6 One request, end to end

Form submit (or file upload → parallel ML NER) - buildPayload() structures the JSON - POST /api/patient/analyze?top_k - build_profile() normalises it into PatientProfile - trial_retrieval pre-filters/annotates all 84 trials - embed_texts() embeds the profile and each trial's criteria (cached) - cosine similarity ranks them - ThreadPool: evaluate_eligibility() per trial (+ Groq review on the top-15, cache-aware) - verdicts merge - AnalyzeResponse returns - the Results page renders verdict cards with reasons and the CSV download. Typically ≈10-20 s for a full 84-trial sweep per patient on CPU when the LLM quota is healthy.

7. What can be improved next

Domain-specific NER. Swap or ensemble DistilBERT-QA with BioBERT / PubMedBERT fine-tuned on CTRI criteria for higher recall on rare entities and lab names.

Dose-threshold rules. The rule engine can parse 'HbA1c <= 9.0' but not yet 'metformin at least 1500 mg/day'; the dose field on medications makes that check possible.

Live registry sync. The Trial Database page imports ClinicalTrials.gov on demand today; a scheduled nightly sync + CTRI export ingestion would keep 84 → thousands of trials fresh automatically.

Doctor feedback loop. Human-in-the-loop corrections that feed back into the rules and the evaluation set - the natural next step to lift recall on the eligible class.

More languages + speech. Hindi notes ship today (13 scripts detected, translated before extraction); regional speech-to-text dictation and ICD / LOINC coding come next.

Accounts + investigator dashboard. Save patient histories, track which trials were contacted, record enrolment outcomes - turning CTQ from a screener into a workflow tool.

PostgreSQL + background jobs. Swap SQLite for PostgreSQL and move long cohort/evaluation runs into background jobs with a progress bar, so refreshing the page never loses a run.

8. Deployment - live on Render

CTQ runs publicly at <https://ctq-clinical-trial-qualifier.onrender.com> (Docker deploy, auto-redeploys on every push to master). To fit the free tier's 512 MB RAM, the DistilBERT NER is exported to ONNX int8 at Docker build time (66 MB self-contained artifact; onnxruntime with single-thread execution and no arena growth - torch is never imported on that path), and embeddings run in a deterministic hashed mode. The Groq key is injected as a sync:false environment variable. Known free-tier limits: after 15 minutes idle the server sleeps (~50 s cold start on the next request), CPU throttling means very large multi-note uploads may time out, and the Groq daily token cap (~200k TPD) occasionally pauses LLM translation/review until the breaker resets - the rule engine keeps verdicts flowing either way.

CTQ - Clinical Trial Qualifier. Backend: FastAPI + SQLite + DistilBERT extractive-QA ML NER + Groq translation/review + sentence-transformers + RapidOCR. Frontend: React + Vite. Deployed on Render. Trial metadata modelled on the Clinical Trials Registry - India (CTRI) with live ClinicalTrials.gov imports. All demo data is synthetic; no

real patient data is used.